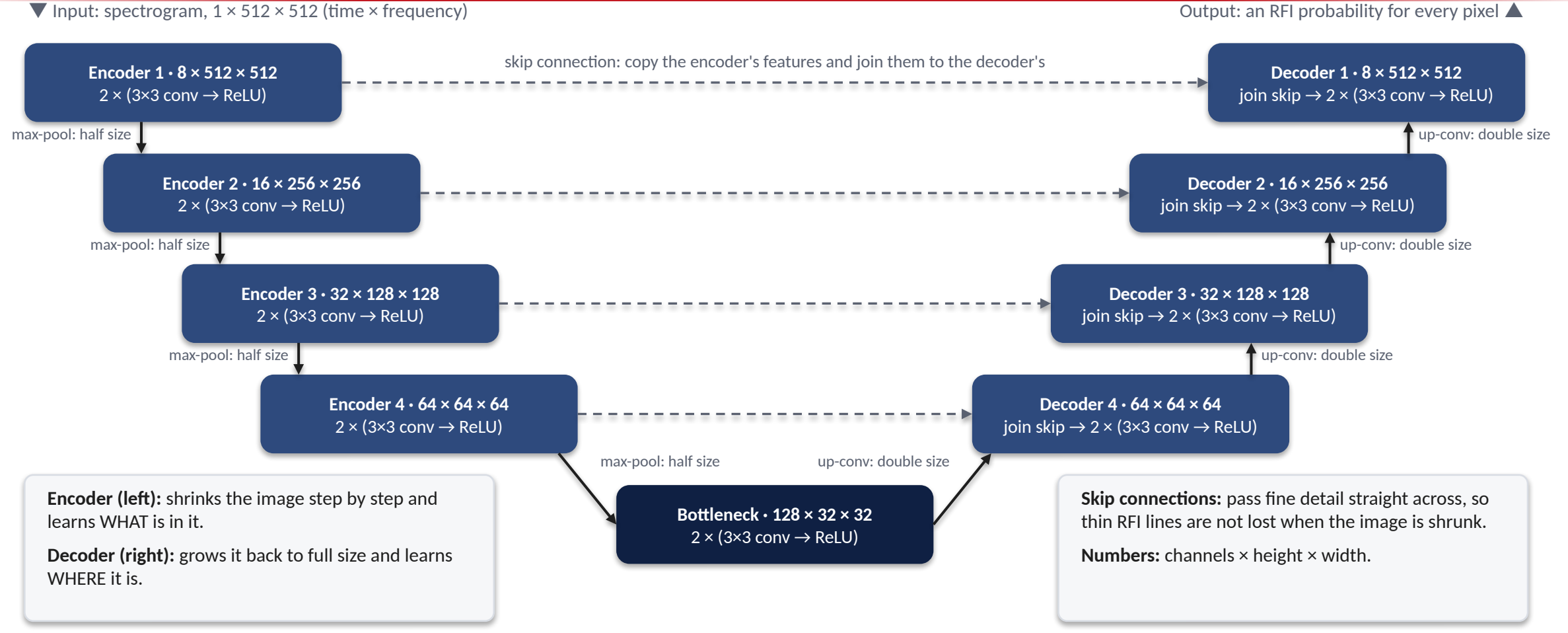

▼ Input: spectrogram, 1 × 512 × 512 (time × frequency)
Output: an RFI probability for every pixel ▲
Encoder 1 · 8 × 512 × 512
2 × (3×3 conv → ReLU)
skip connection: copy the encoder's features and join them to the decoder's
Decoder 1 · 8 × 512 × 512
join skip → 2 × (3×3 conv → ReLU)
max-pool: half size
up-conv: double size
Encoder 2 · 16 × 256 × 256
2 × (3×3 conv → ReLU)
Decoder 2 · 16 × 256 × 256
join skip → 2 × (3×3 conv → ReLU)
max-pool: half size
up-conv: double size
Encoder 3 · 32 × 128 × 128
2 × (3×3 conv → ReLU)
Decoder 3 · 32 × 128 × 128
join skip → 2 × (3×3 conv → ReLU)
max-pool: half size
up-conv: double size
Encoder 4 · 64 × 64 × 64
2 × (3×3 conv → ReLU)
Decoder 4 · 64 × 64 × 64
join skip → 2 × (3×3 conv → ReLU)
max-pool: half size
up-conv: double size
Bottleneck · 128 × 32 × 32
2 × (3×3 conv → ReLU)
Encoder (left): shrinks the image step by step and learns WHAT is in it.
Decoder (right): grows it back to full size and learns WHERE it is.
Skip connections: pass fine detail straight across, so thin RFI lines are not lost when the image is shrunk.
Numbers: channels × height × width.